

# Paul M. Khoury

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## EDUCATION

**Georgia Institute of Technology** - Atlanta, GA

*Aug. 2026 - Present*

M.S. Mechanical Engineering

**Rensselaer Polytechnic Institute** - Troy, NY

*Aug. 2022 - May 2026*

B.S. Mechanical Engineering (Minor: Cognitive Science) GPA: 3.74/4.00; *Magna Cum Laude*; Dean's List; Founder's Award

## WORK EXPERIENCE

**b-ONE Ortho** - Fairfield, NJ

*May 2026 - Aug. 2026*

*R&D Engineer Intern*

- Designed surgical instrument components; applied **GD&T** and advanced drawings toward manufacturing release
- Supported **verification and validation** of pre-release orthopedic implants through analysis, FEA, and physical testing
- Communicated effectively with **international** teams to coordinate design changes and manufacturing workflow updates
- Authored test reports synthesizing lab data and FDA feedback supporting **510(k)** clearance of **orthopedic** implants

**Limbitless Solutions Inc.** - Orlando, FL

*May 2025 - Aug. 2025*

*R&D Intern and NSF Research Scholar*

- Designed and prototyped a wearable vibrotactile array to provide **intuitive sensory feedback** for **pediatric prosthetic** arm users during functional tasks, conveying an **artificial sense of touch**
- Established protocol to test patient localization of vibrotactile stimuli; secured **IRB** approval for **human subject research**
- Selected for undergraduate poster presentation for the 2025 BMES Annual Meeting; co-authored internal paper

**Medline, NAMIC Division** - Glens Falls, NY

*Aug. 2024 - Dec. 2024*

*Manufacturing Engineer Co-op*

- Designed and built a Cartesian **6-axis robot** for precision fluid dispensing in adhesive bonding processes
- Undertook **corrective and preventive action** (CAPA) to significantly reduce particulate contamination in cleanrooms to meet stringent **medical device regulatory standards** in the US, EU, and Japan
- Enhanced **workplace safety** on production floor with custom dividers, fixtures, and light curtains around machinery
- Integrated actuator components into a fully automated production line and programmed them using **PLC ladder logic**
- Reverse-engineered aging machinery and produced detailed part designs, adhering to internal **document** standards

**RPI Lab for Accelerated Computing in Engineering and Science** - Troy, NY

*Jan. 2024 - Jan. 2025*

*Undergraduate Researcher*

- Conducted **finite element modeling** of pelvic fractures to evaluate efficacy of **orthopaedic surgical fixtures**
- Collaborated with a physician to improve upon existing **biomechanical** models

**Zalpha** (formerly 'Invest-ed') - New York, NY

*Nov. 2021 - Aug. 2023*

*Product Team Lead*

- Contributed to the early development of an ed-tech **startup** that delivered an innovative paper trading **mobile app**
- Spearheaded **Product** and **UX design**, collaborating with developers and stakeholders to refine the app's interface

## PROJECTS

**"Thread the Loop" Surgery Skill with Robot arm**

*Sept. 2025 - Present*

- Implemented inverse-kinematics-based control of a 6-DOF robotic arm for maneuvers relevant to laparoscopic surgery.

**EMG Classification using Machine Learning**

*Jan. 2025 - May 2025*

- Developed a classification model to infer gestures based on noisy sEMG signals collected from the wrist/forearm.

**Historical Facade Restoration**

*Sept. 2024 - May 2025*

- Digitally reconstructed and manufactured decorative facade elements for a fire-damaged historic residence.

## LEADERSHIP & VOLUNTEERING

Former RPI Undergraduate Council President; **FIRST Robotics Mentor**; RPI **Biomedical Engineering Society** Project Team; Volunteer Dispatcher (BRAVO Volunteer Ambulance, Brooklyn, NY); Lector at RPI University Chapel + Cultural Center

## SKILLS & CERTIFICATIONS

**ROS**; **GD&T**; **Siemens NX**, **AutoCAD Certified User**; **SOLIDWORKS** CAD Design Associate (CSWA); Six Sigma Green Belt; **LabVIEW**; **Matlab** and **Simulink**; PLC Programming, LTSpice; Microsoft **Excel**; Office 365 Suite; Abaqus **FEA**; HyperMesh **CFD**; **Machine Learning** Engineering with **Python**; **Conversational proficiency** in Polish, Russian; Research & Ethics Training